

Decomposition of Chiral-Symmetric operator

Here we demonstrate that any operator Q satisfying:

$$SQS^{-1} = -Q, \quad S^\dagger = S, \quad S^2 = 1$$

can be written as:

$$Q = S_+QS_- + S_-QS_+,$$

where $S = S_+ - S_-$ is the spectral decomposition of S . Noticing that:

$$Q = (S_+ + S_-)Q(S_+ + S_-),$$

we only need to show that $S_\pm QS_\pm = 0$. And indeed:

$$S_+QS_+ = -S_+(SQS)S_+ = -(S_+S)Q(SS_+) = -S_+QS_+,$$

and similarly for S_-QS_- . The affirmation then follows. Another immediate property is:

$$(S_\pm QS_\mp)(S_\mp Q^{-1}S_\pm) = S_\pm,$$

which we also used in the text. The identity follows by observing that $S_\pm QS_\mp = S_\pm Q$ and that $S_\mp Q^{-1}S_\pm = Q^{-1}S_\pm$, in which case:

$$(S_\pm QS_\mp)(S_\mp Q^{-1}S_\pm) = S_\pm QQ^{-1}S_\pm = S_\pm.$$

Proof of Polarization Relation

Since $\text{Tr}[A(k)] = i \sum_{\alpha \in \text{occ.}} \langle u_\alpha(k) | \partial_k | u_\alpha(k) \rangle = i \sum_{\alpha \in \text{occ.}} \langle Su_\alpha(k) | \partial_k | Su_\alpha(k) \rangle$ the polarization of the occupied bands P_o is equal to that of the unoccupied bands P_u . Also, if we include all the bands then polarization is equal to an integer, and thus, we know that $P_o + P_u \in \mathbb{Z}$, and thus $2P_o \in \mathbb{Z}$. This means that chiral symmetry quantizes the polarization in units of $1/2$, i.e., $P_o = \frac{n}{2}$ for $n \in \mathbb{Z}$.

Now let us consider a basis for a generic Bloch Hamiltonian $h(k)$ such that the chiral operator S is diagonal. To be chiral symmetric, and gapped, $h(k)$ must have an even number of bands; half above zero energy and half below. Thus, when S is diagonalized it generically takes the form $S = \tau^z \otimes \mathbb{I}_N$ where the total number of bands is $2N$, $\mathbb{I}_N$ is the $N \times N$ identity matrix, and τ^z is the diagonal Pauli matrix. With this choice of basis we can choose the Bloch functions of the occupied bands to be of the form

$$|u_\alpha(k)\rangle = \begin{pmatrix} v_{1\alpha}(k) \\ v_{2\alpha}(k) \end{pmatrix} \quad (9)$$

where $\alpha = 1, 2, \dots, N$ and $v_{1\alpha}, v_{2\alpha}$ are N -component spinors. The unoccupied bands can then be written as

$$|Su_\alpha(k)\rangle = \begin{pmatrix} v_{1\alpha}(k) \\ -v_{2\alpha}(k) \end{pmatrix}. \quad (10)$$

To satisfy normalization and the orthogonality of different bands we must have the constraints

$$\begin{aligned} v_{1\alpha}^\dagger(k)v_{1\beta}(k) + v_{2\alpha}^\dagger(k)v_{2\beta}(k) &= \delta_{\alpha\beta} \\ v_{1\alpha}^\dagger(k)v_{1\beta}(k) - v_{2\alpha}^\dagger(k)v_{2\beta}(k) &= 0 \end{aligned} \quad (11)$$

where the second constraint comes from $\langle Su_\alpha(k) | u_\beta(k) \rangle = 0$.

Using this decomposition we can write twice the charge polarization of the occupied bands as

$$2P_o = \frac{2i}{2\pi} \int dk \sum_{\alpha \in \text{occ.}} \left[v_{1\alpha}^\dagger(k) \partial_k v_{1\alpha}(k) + v_{2\alpha}^\dagger(k) \partial_k v_{2\alpha}(k) \right]. \quad (12)$$

The winding number can be written

$$\nu = \frac{i}{\pi} \int dk \sum_{\alpha \in \text{occ.}} \left[v_{1\alpha}^\dagger(k) \partial_k v_{1\alpha}(k) - v_{2\alpha}^\dagger(k) \partial_k v_{2\alpha}(k) \right]. \quad (13)$$

From the constraint that both $2P_o$ and ν are integers we know that $2P_o \pm \nu$ are also integers. This allows us to define

$$\begin{aligned} 2P_o + \nu &\equiv c_+ = \frac{4i}{2\pi} \int dk \sum_{\alpha \in \text{occ.}} \left[v_{1\alpha}^\dagger(k) \partial_k v_{1\alpha}(k) \right] \\ 2P_o - \nu &\equiv c_- = \frac{4i}{2\pi} \int dk \sum_{\alpha \in \text{occ.}} \left[v_{2\alpha}^\dagger(k) \partial_k v_{2\alpha}(k) \right]. \end{aligned} \quad (14)$$

From the definitions of $c_\pm$ we can see that

$$2P_o = \frac{c_+ + c_-}{2}, \quad \nu = \frac{c_+ - c_-}{2}. \quad (15)$$

For $2P_o, \nu$ to be integers $c_\pm$ must be both even or both odd. If they are both even then we have proven

$$2P_o = \nu \pmod{2} \quad (16)$$

since in that case $2P_o = a + b$ and $\nu = a - b$ for integers a, b and the sum and difference of two integers has the same parity. If both $c_\pm$ are odd then the result is

$$2P_o = (\nu + 1) \pmod{2}$$

which means the polarization and winding have an opposite parity relationship. The final step is thus to prove that $c_\pm$ must always be even.

To show this we note that $\sqrt{2}v_{1\alpha}(k)$ and $\sqrt{2}v_{2\alpha}(k)$ are normalized to unity from the constraints in Eq. 11. Thus, for each α we must have $\frac{i}{2\pi} \int dk \left[\sqrt{2}v_{1\alpha}^\dagger \partial_k \sqrt{2}v_{1\alpha} \right]$ and $\frac{i}{2\pi} \int dk \left[\sqrt{2}v_{2\alpha}^\dagger \partial_k \sqrt{2}v_{2\alpha} \right]$ both equal to integers. And finally, this shows that $c_\pm$ are even integers since they are sums over integers (one integer for each α that are subsequently multiplied by a global factor of two which makes the final result an even integer. This leaves us with Eq. 16 as the correct relation between the polarization and winding number.

As a numerical check of the relation between the winding and the polarization of the system, we compute the polarization as a function of the mass parameter and the on-site disorder strength. This can be achieved by computing the eigenvalues of the projected position operator $X_{\mathcal{P}} = \mathcal{P}X\mathcal{P}$, where $\mathcal{P}$ is the projection operator of the occupied states. In a finite system with periodic boundary conditions, the position operator is written in the exponential form $X = \sum_R e^{\frac{2\pi i R}{N}} |R\rangle\langle R|$. Using the set of eigenvalues $\{\xi_n\}$ of $X_{\mathcal{P}}$, the polarization is then given by

$$P_0 = \sum_n \left(\frac{1}{2\pi} \text{Im} \log \xi_n - \frac{n}{N} \right) \quad (17)$$

which just measures the relative shift of the Wannier centers with respect to the lattice sites. In Fig.3, we show the result of this calculation in the m vs. W parameter space, for the case when $W_1 = 0$ and $W_2 = W$. This calculation was obtained by disorder-averaging over 500 realizations. The light-colored region corresponds a $1/2$ polarization and matches the calculation of the ν invariant as well as what is expected from the calculation of the localization length that is discussed in the main text.

Derivation of Critical Scaling

Eq. 6 can also enable us to determine the critical exponent for the transition. Let (m_c, W_1^c, W_2^c) be a point on $\mathcal{S}_c$. We can cross $\mathcal{S}_c$ by varying any of the three parameters, so let us vary m in a small interval $[m_c - \epsilon, m_c + \epsilon]$. Note that Eq. 6 can be rewritten as:

$$\begin{aligned} \Lambda^{-1}(m) &= \left| \ln \left[|2 + W_1^c| \frac{1}{W_1^c} + \frac{1}{2} \right] / |2 - W_1^c| \frac{1}{W_1^c} - \frac{1}{2} \right| \\ &+ \left(\frac{m}{W_2^c} - \frac{1}{2} \right) \ln |2m - W_2^c| - \left(\frac{m}{W_2^c} + \frac{1}{2} \right) \ln |2m + W_2^c|. \end{aligned} \quad (18)$$

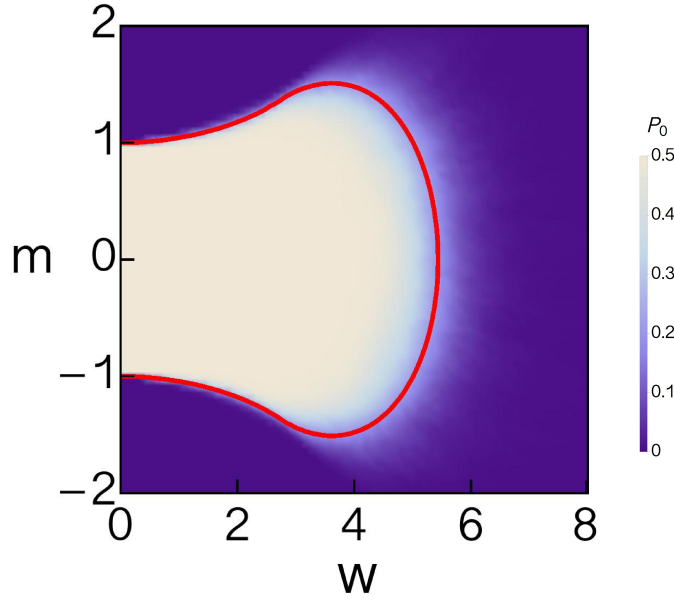


FIG. 3. Polarization of the system disorder-averaged over 500 realizations for $W_1 = 0$ and $W_2 = W$. The light-colored region corresponds to a $1/2$ polarization. This region matches well with the phase boundary determined by the divergence of the localization length, which is denoted in here by the red solid line.

One can see now explicitly that, except for the special cases when $m_c = \pm 1/2 W_2^c$, the function inside the absolute value is analytic of m around m_c , and this function cancels when $m = m_c$. As such, $\Lambda^{-1}(m) = |m - m_c| [c_0 + c_1(m - m_c)^2 \dots]$, which gives a critical exponent 1.

The special critical cases $m_c = \pm 1/2 W_2^c$, marked with line (1) in Fig. 2, occur when the amplitudes of the random and non-random components of the onsite potential are equal. If $m_c = 1/2 W_2^c$:

$$\Lambda^{-1}(m) = \left| \ln \left[|2 + W_1^c|^{\frac{1}{W_1^c} + \frac{1}{2}} / |2 - W_1^c|^{\frac{1}{W_1^c} - \frac{1}{2}} \right] + \frac{1}{W_2^c} (m - m_c) \ln |2(m - m_c)| - \frac{1}{W_2^c} (m + m_c) \ln |2(m + m_c)| \right| \quad (19)$$

The function inside the absolute value sign cancels when $m = m_c$. Since the second term becomes null when $m = m_c$, the last term must cancel the first term when $m = m_c$, and this cancelation occurs in analytic fashion. As such, the behavior of $\Lambda^{-1}(m)$ around m_c is

$$\Lambda^{-1}(m) = \left| \frac{1}{W_2^c} |m - m_c| \ln |m - m_c| + c_1 |m - m_c| + c_2 (m - m_c)^2 \dots \right|,$$

hence the localization length diverges as:

$$\Lambda(m) \sim \frac{W_2^c}{|m - m_c| \ln |m - m_c|}.$$

There is a second class of special critical points, given by $W_1^c = 2$ and marked by line (2) in Fig. 2, where a similar anomalous scaling occurs.

Proof of the simplified form winding number from the spin representation

As shown in the text, the ground state of the random spin-1/2 XX model is formed by singlets of varying lengths. Because of the no-crossing constraint on the singlets every singlet state in the ground state will be formed by one spin

belonging to sublattice A and another spin belonging to sublattice B . Let us associate to the i -th singlet the pair of numbers $d_i = \{d_{i1}, d_{i2}\}$ which are the lattice sites of the two spins involved in forming the singlet. Using this notation, the ground state can be written as

$$\begin{aligned} |\Omega\rangle &= \prod_i \frac{1}{\sqrt{2}} [S_{2d_{i1}}^\dagger - S_{2d_{i2}-1}^\dagger] |\downarrow \dots \downarrow\rangle \\ &= \prod_i \frac{1}{\sqrt{2}} [(-i)^{d_{i1}+1} c_{d_{i1},A}^\dagger \tilde{K}(d_{i1}, 1) - (-i)^{d_{i2}} c_{d_{i2},B}^\dagger \tilde{K}(d_{i2}, 0)] |0\rangle \end{aligned} \quad (20)$$

where $\tilde{K}(m, \lambda) = e^{i\pi\lambda\hat{n}_{m,B}} \exp[i\pi \sum_{j=1}^{m-1} (\hat{n}_{jA} + \hat{n}_{jB})]$. Due to the non-crossing nature of the singlets, the ground state can be simplified to

$$|\Omega\rangle = \prod_i \frac{1}{\sqrt{2}} [\alpha_i c_{d_{i1},A}^\dagger - \beta_i c_{d_{i2},B}^\dagger] |0\rangle \quad (21)$$

where α_i and β_i have unit modulus and their particular values depend on the configuration of the singlets. Let us take this on faith for a moment, we will prove it below after we finish the proof of the simplified winding number formula in this representation of the ground state.

This form of the ground state is a Slater determinant constructed from the single-particle states

$$|\psi_i\rangle = \frac{1}{\sqrt{2}} (\alpha_i c_{d_{i1},A}^\dagger - \beta_i c_{d_{i2},B}^\dagger) |0\rangle \quad (22)$$

which means that we can write explicitly $Q_{+-} = \sum_{i=1}^N (-\alpha_i \beta_i^*) |d_{i1}, A\rangle \langle d_{i2}, B|$. The winding formula can thus be simplified in the following manner. First, we note that

$$\begin{aligned} Q_{-+} \hat{X} Q_{+-} &= \sum_{ij} \alpha_i^* \beta_j \alpha_j \beta_i^* |d_{i2}, B\rangle \langle d_{i1}, A| \hat{X} |d_{j1}, A\rangle \langle d_{j2}, B| \\ &= \sum_{ij} \alpha_i^* \beta_j \alpha_j \beta_i^* |d_{i2}, B\rangle d_{i1} \delta_{ij} \langle d_{j2}, B| \\ &= \sum_i d_{i1} |d_{i2}, B\rangle \langle d_{i2}, B|. \end{aligned} \quad (23)$$

Furthermore, we can write the position operator as $\hat{X} = \sum_i d_{i2} |d_{i2}, B\rangle \langle d_{i2}, B|$, which means that we get

$$-(Q_{-+} \hat{X} Q_{+-} - \hat{X}) = \sum_i (d_{i2} - d_{i1}) |d_{i2}, B\rangle \langle d_{i2}, B|. \quad (24)$$

Hence, the winding number can be explicitly written as

$$\begin{aligned} \nu &= -\mathcal{T} (Q_{-+} [\hat{X}, Q_{+-}]) = -\mathcal{T} (Q_{-+} \hat{X} Q_{+-} - \hat{X}) \\ &= \frac{1}{N} \sum_{i=1}^N (d_{i2} - d_{i1}) \end{aligned} \quad (25)$$

This expression is just the sum of vectors connecting the end-points of the singlets. In particular, since the topological phase with no disorder corresponds to singlets forming between nearest neighbors, we have that $(d_{i2} - d_{i1}) = 1$. The invariant then yields $\nu = \frac{1}{N} \sum_i 1 = 1$, as expected. By contrast, in the trivial phase, singlets form between the B and A spins of the *same* site, which implies $\nu = \frac{1}{N} \sum_i 0 = 0$.

To complete the proof we need to show that the singlet ground state can always be written in the form

$$|\Omega\rangle = \prod_{i=1}^N \frac{1}{\sqrt{2}} [\alpha_i c_{d_{i1},A}^\dagger - \beta_i c_{d_{i2},B}^\dagger] |0\rangle \quad (26)$$

where $|\alpha_i| = 1$ and $|\beta_i| = 1$ for all i . We stress from the outset that the particular form of the α_i and β_i is not important for the winding formula, since these factors drop out in the end. Thus, in the following derivation, we will be concerned exclusively with whether these factors have unit modulus.

We will prove Eq. 26 in an inductive manner. Let us define the following three operators which will help to simplify the notation:

$$\begin{aligned} P(F, G) &= \prod_{i=F}^G Z(i), \\ Z(i) &= \frac{1}{\sqrt{2}} \left[(-i)^{d_{i1}+1} c_{d_{i1},A}^\dagger \tilde{K}(d_{i1}, 1) - (-i)^{d_{i2}} c_{d_{i2},B}^\dagger \tilde{K}(d_{i2}, 0) \right], \\ \tilde{Z}(i) &= \frac{1}{\sqrt{2}} \left[\alpha_i c_{d_{i1},A}^\dagger - \beta_i c_{d_{i2},B}^\dagger \right]. \end{aligned}$$

The ground state, in terms of these operators, is given by $|\Omega\rangle = P(1, N)|0\rangle = \prod_{i=1}^N Z(i)|0\rangle$. The objective is then to show that $|\Omega\rangle = \prod_i^N \tilde{Z}(i)|0\rangle$. To simplify the derivation, we will assume that the labeling of the singlets is such that they are organized according to their length, so that the smallest singlet acts on the vacuum first. This can always be done because the factors in $|\Omega\rangle$ (previous to any simplification) all commute. We start by rewriting the ground state as

$$\begin{aligned} |\Omega\rangle &= P(1, N-1)Z(N)|0\rangle \\ &= P(1, N-1) \frac{1}{\sqrt{2}} \left[(-i)^{d_{N,1}+1} c_{d_{N,1},A}^\dagger - (-i)^{d_{N,2}} c_{d_{N,2},B}^\dagger \right] |0\rangle \\ &= P(1, N-1) \tilde{Z}(N)|0\rangle \end{aligned}$$

where we have used that $\tilde{K}(m, \lambda)|0\rangle = |0\rangle$. We now simplify the next factor:

$$|\Omega\rangle = P(1, N-1) \tilde{Z}(N)|0\rangle \quad (27)$$

$$= P(1, N-2) \frac{1}{\sqrt{2}} \left[(-i)^{d_{N-1,1}+1} c_{d_{N-1,1},A}^\dagger \tilde{K}(d_{N-1,1}, 1) - (-i)^{d_{N-1,2}} c_{d_{N-1,2},B}^\dagger \tilde{K}(d_{N-1,2}, 0) \right] \tilde{Z}(N)|0\rangle \quad (28)$$

$$= P(1, N-2) \frac{1}{\sqrt{2}} \left[(-i)^{d_{N-1,1}+1} c_{d_{N-1,1},A}^\dagger \tilde{K}(d_{N-1,1}, 1) \tilde{Z}(N)|0\rangle - (-i)^{d_{N-1,2}} c_{d_{N-1,2},B}^\dagger \tilde{K}(d_{N-1,2}, 0) \tilde{Z}(N)|0\rangle \right]. \quad (29)$$

Because of the non-crossing rule and the way we have organized the singlets according to their length, there are three possible situations:

1. *The interval d_N lies to the right of the interval d_{N-1} :* In this case, neither $\tilde{K}(d_{N-1,2}, 0)$ nor $\tilde{K}(d_{N-1,1}, 1)$ will have number operators in their exponentials with corresponding creation operators in $Z(N)$, which means that the K operators commute with $Z(N)$. Hence, we can write $|\Omega\rangle = P(1, N-2) \tilde{Z}(N-1) \tilde{Z}(N)|0\rangle$.
2. *The interval d_N lies inside of the interval d_{N-1} :* Then one, and only one, of $\tilde{K}(d_{N-1,2}, 0)$ and $\tilde{K}(d_{N-1,1}, 1)$ will have the number operators associated with *both* of the creation operators in $Z(N)$. The state $Z(N)|0\rangle$ is thus an eigenstate of both K operators, with eigenvalues ± 1 , depending on which K has operators shared with $Z(N)$. Hence, $|\Omega\rangle = P(1, N-2) \tilde{Z}(N-1) \tilde{Z}(N)|0\rangle$.
3. *The interval d_N lies to the left of d_{N-1} :* Then both $\tilde{K}(d_{N-1,2}, 0)$ and $\tilde{K}(d_{N-1,1}, 1)$ will have the number operators associated with *both* of the creation operators in $Z(N)$. The state $Z(N)|0\rangle$ is thus an eigenstate of both K operators, with eigenvalues -1 . Hence, $|\Omega\rangle = P(1, N-2) \tilde{Z}(N-1) \tilde{Z}(N)|0\rangle$.

In all three cases, we get the desired simplification of the ground state. This is the first inductive step. For the m -th step of the inductive argument, we now assume that the following expression holds

$$|\Omega\rangle = P(1, N-m) \prod_{i=N}^{N-m+1} \tilde{Z}(i)|0\rangle, \quad (30)$$

and we must now show that this implies the $m+1$ case, namely that

$$|\Omega\rangle = P(1, N-m-1) \prod_{i=N}^{N-m} \tilde{Z}(i)|0\rangle. \quad (31)$$

To show this, we again write

$$|\Omega\rangle = P(1, N-m) \prod_{i=N}^{N-m+1} \tilde{Z}(i)|0\rangle \quad (32)$$

$$= P(1, N-m-1) \frac{1}{\sqrt{2}} \left[(-i)^{d_{N-m,1}+1} c_{d_{N-m,1},A}^\dagger \tilde{K}(d_{N-m,1}, 1) - (-i)^{d_{N-m,2}} c_{d_{N-m,2},B}^\dagger \tilde{K}(d_{N-m,2}, 0) \right] \prod_{i=N}^{N-m+1} \tilde{Z}(i)|0\rangle \quad (33)$$

We can group the factors $\prod_{i=N}^{N-m+1} \tilde{Z}(i) = B_1 B_2 B_3$ in the following way:

- B_1 is made out of singlets with intervals that lie to the right of the interval d_{N-m} : In this case, neither $\tilde{K}(d_{N-m,2}, 0)$ nor $\tilde{K}(d_{N-m,2}, 1)$ will have number operators in their exponentials with corresponding creation operators in B_1 , which means that the $\tilde{K}$ operators commute with B_1 .
- B_2 is made out of singlets with intervals that lie inside of the interval d_{N-m} : Then one, and only one, of $\tilde{K}(d_{N-m,2}, 0)$ and $\tilde{K}(d_{N-m,2}, 1)$ will have all of the number operators associated with the creation operators in B_2 . Let us denote the number of factors in B_2 by ρ_{B_2} .
- B_3 is made out of singlets with intervals such that d_{N-m} lies to the left of them: Then both $\tilde{K}(d_{N-m,2}, 0)$ and $\tilde{K}(d_{N-m,2}, 1)$ will have all of the number operators associated with the creation operators in B_3 . Let us denote the number of factors in B_2 by ρ_{B_3} .

Hence, depending on which case occurs with the B_2 singlets, we will either get

$$|\Omega\rangle = P(1, N-m-1) \frac{(-1)^{n_{B_3}}}{\sqrt{2}} \left[(-i)^{d_{N-m,1}+1} c_{d_{N-m,1},A}^\dagger (-1)^{n_{B_2}} - (-i)^{d_{N-m,2}} c_{d_{N-m,2},B}^\dagger \right] B_1 B_2 B_3 |0\rangle \quad (34)$$

or

$$|\Omega\rangle = P(1, N-m-1) \frac{(-1)^{n_{B_3}}}{\sqrt{2}} \left[(-i)^{d_{N-m,1}+1} c_{d_{N-m,1},A}^\dagger - (-i)^{d_{N-m,2}} c_{d_{N-m,2},B}^\dagger (-1)^{n_{B_2}} \right] B_1 B_2 B_3 |0\rangle \quad (35)$$

In both cases, we conclude that the singlet ground state can be written as

$$|\Omega\rangle = P(1, N-m-1) \prod_{i=N}^{N-m} \tilde{Z}(i)|0\rangle. \quad (36)$$

As a consequence, by successive application of this procedure, we can finally arrive at the desired result

$$|\Omega\rangle = \prod_{i=1}^N \tilde{Z}(i)|0\rangle. \quad (37)$$

Scaling of Entanglement Entropy

Considering the scaling of the entanglement entropy as a function of the partition size has become a common technique for the study of 1d critical points [16,17]. When tuned to the critical point one expects that the entanglement entropy will scale proportional to the log of the partition size with a proportionality constant that includes the central charge of the 1d CFT critical point. For the random-singlet critical point the scaling should be

$$S = \frac{c \log(2)}{3} \log \left[\frac{N}{\pi} \sin \left(\frac{\pi x}{N} \right) \right] + s_0 \quad (38)$$

where s_0 is a non-universal constant and c is the central charge of the CFT associated with the system at criticality in the clean limit, which is $c = 1$ in our case, N is the total length of the system and x is the length of the partition size. The $\log(2)$ factor is a disorder-induced “renormalization” of the central charge that is characteristic for certain disordered spin models [40]. We explored the scaling of the entanglement using the fermionic model at the critical point, and found that it matches Eq. 38. This further supports the notion that the critical point of the chiral symmetric model is that of the random singlet phase. The numerical fit can be found in Fig. 4.

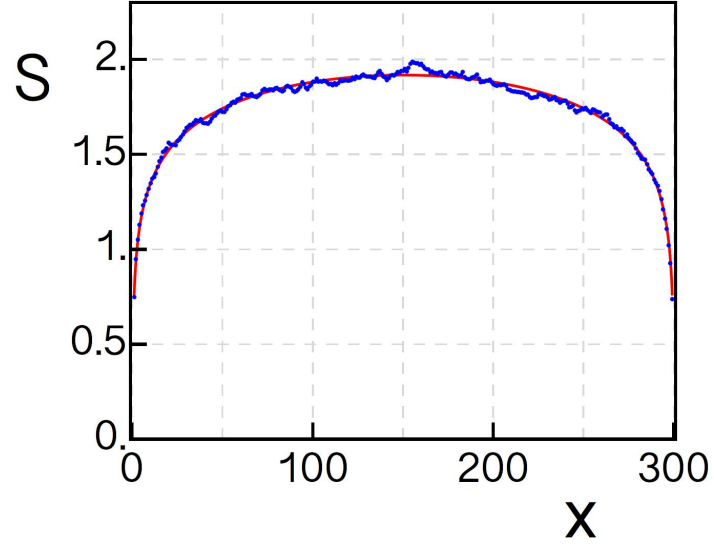


FIG. 4. Scaling of the entanglement entropy of a section of size x when the parameters are tuned to criticality. The blue dots correspond to the numerical calculation, and averaging was done over 200 disorder realizations for a lattice of size $N = 300$. The red solid line corresponds to Eq. 38.